

Domain Research Note

EduAgent: Adaptive Multi-Agent AI Tutor

Background

Recent advancements in Large Language Models (LLMs) have enabled systems that can perform reasoning and interact with external tools. The ReAct framework (Yao et al., 2023) is a key development in this area, combining reasoning (Thought) and tool usage (Action) in a structured loop. This approach reduces hallucination and improves problem-solving by grounding responses in retrieved information.

Problem with Existing Systems

While ReAct is effective for general reasoning tasks, it is not designed for educational use. It treats every user identically and lacks mechanisms required for personalized learning:

- No memory of previous learner interactions
- No adaptation to learner skill level
- Fixed few-shot examples regardless of context
- No evaluation of learner understanding
- Single-agent architecture handling all tasks

As a result, such systems behave like generic assistants rather than intelligent tutors.

Research Gap

The key limitation identified is that ReAct-based agents are **stateless and learner-agnostic**. In educational settings, effective tutoring requires:

- identifying learner proficiency,
- adapting explanation depth,
- tracking weak concepts,
- maintaining long-term learning history,
- evaluating comprehension through interaction.

Existing agent frameworks do not integrate these capabilities into a unified system.

Proposed Work

To address this gap, we propose **EduAgent**, an extension of the ReAct framework for adaptive AI tutoring. The system introduces:

- **Difficulty Classifier:** A DistilBERT model trained on a custom dataset to detect learner level (beginner, intermediate, advanced)
- **Dynamic Example Retrieval:** Selection of relevant examples from a dataset instead of fixed prompts
- **Memory Agent:** Persistent learner profile storing history, weak areas, and progress
- **Evaluator Agent:** Generates follow-up questions to assess understanding
- **Multi-Agent Architecture:** Separation of tutoring, evaluation, and memory into distinct components

The system follows an adaptive learning loop:

Question → Classify → Retrieve → Teach → Evaluate → Update Memory

Significance

EduAgent transforms the ReAct framework from a general-purpose reasoning agent into a personalized tutoring system. By incorporating learner-awareness, adaptive feedback, and memory, the system moves closer to real-world educational requirements.

Conclusion

This research identifies a critical gap in existing LLM-based agent systems: the lack of personalization in learning environments. EduAgent addresses this by integrating classification, retrieval, evaluation, and memory into an agentic tutoring pipeline, enabling adaptive and learner-centric AI education.